ETHAN COSKAY

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EDUCATION

Cornell University Ithaca, New York

College of Engineering - B.S. in Computer Science, Minor in Artificial Intelligence *Expected May 2027*

Relevant Coursework: Introduction to Deep Learning, Introduction to Computer Vision, Introduction to Machine Learning, Analysis of Algorithms, Foundations of AI Reasoning and Decision-Making, Foundations of Robotics, AI: Law, Ethics, and Policy, Systems Programming, Object-Oriented Programming and Data Structures, Computer System Organization and Programming, Data Structures and Functional Programming, Probability and Statistics, Discrete Math

TECHNICAL SKILLS

Languages: Python, C++, C, Java, C#, JavaScript, TypeScript, Swift, OCaml

AI/ML: PyTorch, TensorFlow, Scikit-Learn, OpenCV, Machine Learning, Applied AI

Backend & Frameworks: FastAPI, React, Flask, .NET, Node.js, REST APIs

Databases: SQL, MongoDB, Cosmos DB, SQLite

Cloud & DevOps: Docker, Azure, Google Cloud Platform (GCP), CI/CD

Tools: Git, GitHub, VS Code, Postman, ffmpeg, MSTest/Moq

WORK EXPERIENCE

Software Engineer Intern - Microsoft, Redmond, WA *Jun 2026 – August 2026*

- Built a dedup-aware smart tagging system that automatically propagates user-applied tags across duplicate documents in review sets containing up to 5 million items, eliminating manual tagging of duplicate copies
- Engineered nested-ID expansion and case-insensitive merge logic within TagById workflows to enforce consistent tagging across document families and duplicate groups
- Integrated SearchCore with CFS embeddings and similarity orchestration, backed by Cosmos DB and cache-based embedding reuse, to power AI-assisted tag recommendations across review sets — saving users an estimated 1,500+ hours of manual document tagging
- Designed deterministic fallback behavior for incomplete or missing dedup data, preventing downstream pipeline failures and improving production resilience
- Reduced feature-rollout risk using ECS flight gating, enabling controlled activation and zero-downtime rollback

Tech: *C#, .NET, SearchCore, ReviewSetCore, ECS Flighting, CFS, Cosmos DB, Distributed Caching, MSTest/Moq*

Software Engineer Intern - Davidson Kempner Capital Management, New York, NY *May 2025 – May 2026*

- Built a Python code generator that auto-produced Enums, request/response objects, and POPOs from T4 templates, eliminating 5,000+ lines of manual boilerplate per code-generation cycle
- Architected a full client-server layer, including a custom JSON deserializer and HTTP client processor enabling Python developers to query REST APIs and backend databases directly, unblocking end-to-end quantitative research and data-access workflows
- Designed custom code sections that persist user modifications across code regenerations, reducing rework and maintenance overhead for the engineering team
- Built a configuration service that dynamically routed topics to the correct servers, improving reliability and fault tolerance of distributed API requests

Tech: *C#, Python, T4 Templates, REST APIs, JSON, HTTP Client Architecture, System Design*

Software Engineer Intern - LinkedIn, Ithaca, NY *Sep 2024 – May 2025*

- Built interactive front-end components, including modal-based interactions and video integration, for a LinkedIn-style homepage prototype using React and Ant Design
- Partnered directly with a cross-functional team of designers to align design and implementation and ship core features on schedule
- Delivered a polished MVP presented live at LinkedIn's NYC headquarters

Tech: *React, Ant Design, JavaScript, TypeScript*

Software Engineer Intern - Helmes, Vilnius, Lithuania *Jun 2024 – Aug 2024*

- Cleaned and geographically clustered client data using SQLite for the NitroGeo API, improving data usability for downstream analysis
- Migrated NitroGeo from .NET 6 to .NET 8 with zero breaking changes, extending the API's supported lifecycle
- Audited project dependencies and verified NuGet package compatibility, preventing deployment and compatibility issues

Tech: *C#, .NET 6/8, SQLite, NuGet*

PROJECT TEAM EXPERIENCE

Software Team Lead - Cornell Engineering - XR Project Team, Ithaca, NY *Mar 2024 – Present*

- Led a team of 5 engineers building a real-time OCR pipeline for a real-time document-scanning application on Meta Glasses that converts scanned images directly into formatted LaTeX files, enabling structured document conversion
- Designed and manage a MongoDB database schema optimized for storing and querying scanned images, improving data retrieval speed
- Improved text-extraction accuracy through targeted enhancements to document recognition algorithms using Python and OpenCV

Tech: *Python, OpenCV, MongoDB, OCR, LaTeX*

PERSONAL PROJECTS

Investment Intelligence System - github.com/EcSky19/InvestmentIntelligenceSystem *Nov 2025 – Present*

- Architected a modular investment intelligence system integrating six analytical engines; trend detection, fundamentals, institutional flow, technical signals, tape analysis, and sentiment modeling for automated stock evaluation
- Built a Technical Signals Engine and Tape Analysis Engine that detect technical setups and institutional trading activity, generating structured signals, confidence scores, and scenario-based trade implications
- Designing an AI agent layer to synthesize outputs across all six engines, resolve conflicting signals, and produce probabilistic, explainable market intelligence reports

Tech: *Python, financial data APIs, signal processing, agent orchestration*

OneCompute - Microsoft Intern Week Hackathon - 3rd Place Award | 1200+ participants *Summer 2026*

- Owned end-to-end demo and product experience for a privacy-conscious distributed compute platform that turns underutilized employee devices into an opt-in compute network, as the primary judged artifact for a 3-person, 2-day build
- Built a FastAPI-based orchestrator with signed, sandboxed job execution, a native Windows desktop client with custom taskbar and system tray integration, and a live dashboard for real-time monitoring of devices, workloads, and compute capacity
- Directed the demo video, engineering timed zoom effects synced to live UI events via ffmpeg, and validated the full stack across 90 source files and 159 automated tests to deliver a judge-ready 4–5 minute product story

Tech: *Python, FastAPI, Windows Desktop, ffmpeg, Automated Testing*

LEADERSHIP & ACTIVITIES

- Teaching Assistant, CS 1110 - Cornell Engineering: Prepared weekly lesson plans and mentored a class of 20 students in introductory computer science and Python fundamentals
- Successfully completed the selective, 1-year **Selander Center for Engineering Leadership Program** at Cornell Engineering.
- Vice President of Communications, Alpha Sigma Phi - Iota Chapter